



Flowers of the fly orchid



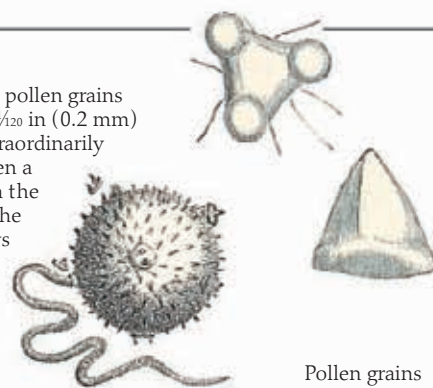
The real thing!

Female impersonators

Some orchids use a clever trick to ensure that they are pollinated. Each flower looks and smells like a female fly, wasp, or bee. The impersonation is so convincing that the male tries to mate with the flower. When this happens, pollen sticks to the bee's body, and when the insect eventually flies off, the pollen is carried to the next flower.

POLLEN GRAINS

Although the largest pollen grains measure only about $\frac{1}{120}$ in (0.2 mm) across, they have extraordinarily intricate shapes. When a pollen grain lands on the stigma of a plant of the same species, it grows a fine tube down the style (p. 17) until it reaches the ovule, or female cell (pp. 26–27).



Pollen grains magnified many times to show the variety of shapes



Long proboscis



Short proboscis

BUTTERFLY POLLINATION

Butterflies are also important pollinators but, unlike bees, they do not feed on the pollen and so do not actively collect it. Instead, when they land on a flower to feed on the nectar, pollen from the anthers sticks to their bodies, ready to be carried to the next flower. Because butterflies have a highly developed sense of smell, butterfly-pollinated flowers are often scented. Many flower in late summer, when butterflies are most abundant.

Proboscis sucks up nectar

DRINKING STRAWS

Butterflies and moths suck up nectar through the proboscis, which is hollow like a drinking straw. The proboscis may vary in length from a fraction of a millimeter to 1 ft (30 cm). When resting, it is coiled up under the butterfly's head.



Pollen from anthers sticks to the butterfly's body

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